

A COUNTEREXAMPLE TO THE UNRESTRICTED DARGAD–LARSSON DOMINATION PROBLEM

ABSTRACT. Dargad and Larsson asked in Problem 7.1 whether their Proposition 7.2 accounts for all dominated options in truncated-support games at the knife’s edge. We determine the complete order among the periodic game values. Writing

$$G_n = \mathcal{T}_{b-a}(n; a, b), \quad q = b - a + 1,$$

we prove, for $0 \leq r, s \leq b$, that

$$G_r \leq G_s \iff \left\lfloor \frac{r}{q} \right\rfloor = \left\lfloor \frac{s}{q} \right\rfloor \text{ and } r \leq s.$$

Thus the values form increasing chains on consecutive q -blocks and are incomparable across blocks. This confirms the contextual reading of Problem 7.1 for the $b+1$ representative positions. Under its unrestricted wording, the comparisons in the source are complete if and only if $q \mid (a-1)$. Otherwise the statement is false: for $(a, b) = (3, 5)$, the Left option G_4 of G_6 is dominated by G_5 , a comparison outside the range of Proposition 7.2.

1. INTRODUCTION

In a partizan subtraction game, Left and Right remove tokens from a single heap according to different subtraction sets; see [1]. Dargad and Larsson introduced the truncated-support game $\mathcal{T}_\tau(n; a, b)$, in which

$$S_L = \{1, \dots, a\}, \quad S_R = \{\tau + 1, \dots, b\}, \quad 0 < a < b.$$

At the knife-edge truncation $\kappa = b - a$, they proved the game-value periodicity

$$(1) \quad \mathcal{T}_\kappa(n; a, b) = \mathcal{T}_\kappa(t; a, b), \quad t \equiv n \pmod{b+1}, \quad 0 \leq t \leq b,$$

and asked whether the only dominated options are those given by their Proposition 7.2 [2, Problem 7.1]; that proposition is stated there under the hypothesis $3 \leq a < b$.

The paragraph preceding that proposition restricts attention to the representatives with heap sizes $0, \dots, b$, whose options have heap sizes at most $b-1$. This is why the comparison range of Proposition 7.2, quoted at the end of this section, ends at $b-2$: the last consecutive pair of option heap sizes available to a representative is $(b-2, b-1)$. Heap size $b+1$ is the first position with an option of heap size b , and the residue b is the one the

2020 *Mathematics Subject Classification.* 91A46, 91A05.

Key words and phrases. partizan subtraction game, truncated support, dominated option, game order.

source's comparisons never reach. The printed question itself contains no restriction on n . We resolve both readings. The structural result is the exact partial order on the $b + 1$ periodic values; the unrestricted reading then has the counterexample stated in the abstract.

Put

$$q = b - a + 1, \quad G_n = \mathcal{T}_{b-a}(n; a, b).$$

At the knife edge, Left removes an element of $\{1, \dots, a\}$ and Right removes an element of $\{q, \dots, b\}$. Hence, for $0 \leq n \leq b$,

$$(2) \quad G_n = \{G_{n-i} : 1 \leq i \leq \min(a, n) \mid G_{n-j} : q \leq j \leq \min(b, n)\}.$$

Statements from the source. We reproduce the three statements of [2] on which the discussion below turns; the only changes are the symbol $\mathcal{T}$ for the ruleset and ordinary set notation. Observation 5.2 reads: “Let a, b and τ be positive integers such that $a < b$ and $\tau < b$. If $n \leq \tau$, then Right does not have any move, whereas Left can remove tokens one by one. Thus, for all $1 \leq n \leq \tau$, $\mathcal{T}_\tau(n; a, b) = n$.” Proposition 7.2 reads: “Consider $G = \mathcal{T}_\kappa(n; a, b)$ and $H = \mathcal{T}_\kappa(n + 1; a, b)$ where $3 \leq a < b$. Then, $G \leq H$ for all $n \in \{b - a + 1, \dots, b - 3, b - 2\} \setminus Q$ ”, where

$$Q = \{(m + 1)(b - a) + m : m \in \mathbb{Z}_{\geq 1}\} = \{(m + 1)q - 1 : m \geq 1\}$$

is the set introduced in the paragraph preceding it. That paragraph presents the games $\mathcal{T}_\kappa(n; a, b)$ with $0 \leq n \leq b$ as representatives of the canonical forms, and records: “The options of these games have heap sizes ranging from 0 to $b - 1$. Thus, only the games with heap sizes $0, 1, \dots, b - 1$ are required to be compared.” Problem 7.1 reads: “Is it true that the only dominated options of $\mathcal{T}_{b-a}(n; a, b)$ are given by Proposition 7.2.”

2. THE ORDER OF THE PERIODIC VALUES

Theorem 1. *For all $0 \leq r, s \leq b$,*

$$(3) \quad G_r \leq G_s \iff \left\lfloor \frac{r}{q} \right\rfloor = \left\lfloor \frac{s}{q} \right\rfloor \text{ and } r \leq s.$$

Consequently, if

$$B_m = \{mq, mq + 1, \dots, \min((m + 1)q - 1, b)\},$$

then the order on $G_0, \dots, G_b$ is the disjoint union of the strict chains indexed by the nonempty blocks B_m . In particular, these $b + 1$ values are pairwise distinct, and $b + 1$ is the least positive period in (1).

Proof. We use strong induction on $r + s$. The case $r = s = 0$ is immediate. Recall that, for short games, $X \leq Y$ if and only if no Left option X^L satisfies $Y \leq X^L$ and no Right option Y^R satisfies $Y^R \leq X$ [3].

Suppose first that r and s lie in the same q -block and $r \leq s$. Every Left option G_u of G_r has $u < r \leq s$, so the induction hypothesis gives $G_s \not\leq G_u$. Every Right option G_v of G_s satisfies $v \leq s - q$ and therefore lies in a block strictly before the common block of r and s ; in the first block there are no

Right options. Hence $G_v \not\leq G_r$ by induction, and the comparison criterion gives $G_r \leq G_s$.

If r and s lie in the same block and $r > s$, then G_{r-1} is a Left option of G_r , and $r-1$ remains in that block. By induction, $G_s \leq G_{r-1}$, so $G_r \not\leq G_s$.

Now suppose that the block of r is later than the block of s , and set

$$u = \max\{s, r - a\}.$$

Since $r \leq b = a + q - 1$, we have $r - a \leq q - 1$. If $s < q$, then $u < q$; if $s \geq q$, then $u = s$. Thus u lies in the same block as s . Moreover, $r - a \leq u < r$, so G_u is a Left option of G_r . Induction gives $G_s \leq G_u$, and therefore $G_r \not\leq G_s$.

Finally, suppose that the block of r is earlier than the block of s , and set

$$v = \min\{r, s - q\}.$$

Writing $i = \lfloor r/q \rfloor$ and $j = \lfloor s/q \rfloor$, we have $j > i$ and

$$s - q \geq (j - 1)q \geq iq, \quad r < (i + 1)q.$$

Hence v lies in the same block as r . Also $q \leq s - v \leq b$, so G_v is a Right option of G_s . By induction, $G_v \leq G_r$, and therefore $G_r \not\leq G_s$. This proves (3). $\square$

3. DOMINATION AND THE COUNTEREXAMPLE

For a heap size n , reduce each option's target heap size modulo $p = b + 1$ to a residue in $\{0, \dots, b\}$. Distinct options of the same player have distinct residues: the difference between two legal move sizes is at most $a - 1 < p$. Theorem 1 therefore gives the complete domination rule.

Corollary 2. *Among Left options whose target residues lie in one q -block, every option except the one with largest residue is dominated. Among Right options whose target residues lie in one q -block, every option except the one with smallest residue is dominated. No domination occurs between options whose target residues lie in different blocks.*

Throughout, “given by Proposition 7.2” means “derivable from Observation 5.2 and Proposition 7.2 by transitivity and the periodicity (1)”. This is the reading intended by the paragraph preceding Proposition 7.2, and both verdicts below are taken under it. Without that latitude the answer is negative for every $a \geq 2$, and for a trivial reason: the comparisons inside the first block come from Observation 5.2 and not from Proposition 7.2, whose range begins at q .

Recall that $Q = \{(m + 1)q - 1 : m \geq 1\}$ is the exceptional set of Proposition 7.2. For $q \leq n \leq b - 1$, Theorem 1 gives

$$G_n < G_{n+1} \iff n \notin Q;$$

when $n \in Q$, the two values are incomparable. Thus, for $3 \leq a < b$, Proposition 7.2 of [2] supplies all consecutive comparisons required among the option values $G_0, \dots, G_{b-1}$ of the representative positions, while Observation 5.2

supplies the first block. This answers the contextual, representative-position reading of Problem 7.1 affirmatively.

For the unrestricted reading, the omitted residue b is decisive. Since

$$b = q + a - 1,$$

the final block is the singleton $\{b\}$ exactly when $q \mid (a - 1)$. In that case Corollary 2 gives no comparison involving G_b , so Observation 5.2 and Proposition 7.2, together with transitivity and periodicity, account for every dominated option at every heap size.

If $q \nmid (a - 1)$, then $b - 1$ and b lie in the same block, and Theorem 1 gives

$$G_{b-1} < G_b.$$

Since $q \mid 0$, the hypothesis $q \nmid (a - 1)$ forces $a \geq 2$, so both values are Left options of G_{b+1} , reached by removing 2 and 1 tokens, respectively. Hence the option G_{b-1} is dominated by G_b . The comparison occurs at index $b - 1$, outside the printed range $n \leq b - 2$ of Proposition 7.2. Thus, under the convention above, the unrestricted statement holds if and only if $q \mid (a - 1)$.

The two values of a outside the hypothesis of Proposition 7.2 agree with this. For $a = 1$ each player has a single legal move size, so no position has two options of the same player and nothing is dominated; the statement is true, and indeed $q = b$ divides $a - 1 = 0$. For $a = 2$ the range $\{q, \dots, b - 2\}$ of Proposition 7.2 is empty, so the proposition supplies nothing, while Observation 5.2 covers exactly the first block $B_0 = \{0, \dots, b - 2\}$; since $q = b - 1$ does not divide $a - 1 = 1$, the comparison $G_{b-1} < G_b$ is again unsupplied and the statement is false.

In particular, for $(a, b) = (3, 5)$ one has $q = 3$ and

$$G_3 < G_4 < G_5.$$

At heap size 6, Left may move to G_5, G_4 , or G_3 , so G_4 is dominated by G_5 . Proposition 7.2 supplies only the comparison $G_3 \leq G_4$ for this ruleset. Thus the unrestricted form of Dargad and Larsson's Problem 7.1 is false.

REFERENCES

- [1] E. Duchêne, M. Heinrich, R. J. Nowakowski, and A. Parreau, *Partizan subtraction games*, *Integers* **21B** (2021), Paper A8.
- [2] A. Dargad and U. Larsson, *Partizan subtraction with full and truncated support*, arXiv:2607.27989v1 [math.CO], 2026.
- [3] A. N. Siegel, *Combinatorial Game Theory*, Graduate Studies in Mathematics, vol. 146, American Mathematical Society, Providence, RI, 2013.